RAMAIAH Institute of Technology

Dept. of Computer Science and Engineering (AI & ML) & Computer Science and Engineering (Cyber Security)

Programme: B E – Computer Science and Engineering (AI&ML) Computer Science and Engineering (Cyber Security)

Internal Assessment – I

TERM : NOV 2023 to MARCH 2024	COURSE NAME : Database Management System
DATE : **28-02-2024** TIME: **10.00 AM-11.00AM**	COURSE CODE : CY32/CI32
MAX MARKS: **30**	PORTIONS : 3^{rd} , 4^{th} & 5^{th} Module

Mobile Phones are banned

Instructions to Candidates: **Answer any two full questions.** Marks: **15x2=30**

Q. NO	**Questions**	**Blooms Levels**	**CO**	**Marks**
1.a	Given the set of FD's, F: { A→B, C→B, D → ABC, AC→ D}. Find the minimal cover of F.	**L3**	**CO4**	**7**
1.b	Consider the following tables: STUDENT (<u>regno: string</u>, name: string, major: string, bdate: date) COURSE (<u>course_id:int</u>, cname: string, dept: string) ENROLL (<u>regno: string, course_id:int</u>, sem: int, marks: int) BOOK _ ADOPTION (<u>course_id:int</u>, sem: int, <u>book-ISBN: int</u>) TEXT (<u>book-ISBN: int</u>, book-title: string, publisher: string, author: string) Write SQL queries for the following i) List the students name who have enrolled for 2nd sem. ii) List no of books adopted in semester '4'. iii) List the courses where course name begins with 'a' being held by 'cs' department. iv) List department which has more than 2 courses.	**L3**	**CO3**	**8**
2.a	Consider the relation R (A, B, C, D, E, F, G, H) with the following sets of FD's {A→BD, B→C, E→FG, AE→H} i . Identify the candidate key(s) for R. ii. Identify the highest normal form that R satisfies (1NF, 2NF, 3NF, or BCNF). iii. If R is not in BCNF, decompose it into a set of BCNF relations.	**L3**	**CO4**	**8**
2.b	Consider the following tables: Branch (<u>branch-name</u>, branch-city, assets) Customer (<u>account-number</u>, customer-name, customer-street, customer-city) loan (<u>loan-number</u>, <u>branch-name</u>, amount) Borrower (<u>account-number</u>, customer-name, <u>loan-number</u>) Account (<u>account-number</u>, branch-name, balance) Depositor (customer-name, <u>account-number</u>) Write the SQL Queries for the following i. List in alphabetical order all customers who have a loan at 'MSRIT' branch. ii. Find the names of all customers who area address includes the substring "yesvanthpur". iii. List the loan numbers in ascending order of amount. iv. Find the average loan amount at each branch.	**L3**	**CO3**	**7**
3.a	What is functional dependency? Explain the inference rules for functional dependency with proof.	**L1**	**CO4**	**8**
3.c	With a neat diagram write and explain the state transition diagram illustrating the states for transaction execution.	**L2**	**CO5**	**7**

*** L1** – Remember, **L2** – Understand, **L3-** Apply.

COs	BL mapping to CO	Marks allotted to BL1	Marks allotted to BL2	Marks allotted to BL3	Marks allotted to BL4	Marks allotted to BL5	Marks allotted to BL6
CO3				15			
CO4		8		15			
CO5			7				

CIE QP Review Format

Course Name: Database Management System **Course Code: CY32/CI32**

1. **Is the mapping of question to the Bloom's level specified in the question paper, correct? If not, suggest the appropriate Bloom's level.**

Question No	1a	1b	1c	2a	2b	2c	3a	3b	3c
If not correct, specify L1/L2/L3/L4/L5/L6									

2. **Is the mapping of question to the CO specified correct or not? If not, suggest the modifications, in mapping to correct CO**

Question No	1a	1b	1c	2a	2b	2c	3a	3b	3c
If not correct, specify the correct CO									

Comment on CO Coverage: ____

3. **Your comment on coverage of Bloom's Level in the CIEs**

____

4. **Grade the overall quality of the question paper on a scale of 1 - 5 (1 being the minimum and 5 is the Maximum): __________**

5. **Overall Suggestion: ____**

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Name & Signature of Reviewer-1 **Name & Signature of Reviewer-2**

Signature of the BOE Chairman

Date of Review:

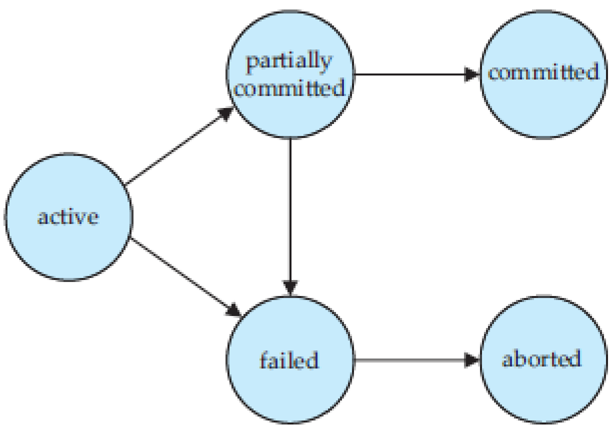
Programme: B E – CSE (AI&ML) & CSE (CY)
Internal Assessment – I

TERM : NOV to MARCH	COURSE NAME : Database Management System
DATE : **03-01-2024** TIME: **3.30-4.30**	COURSE CODE : CY32/CI32
MAX MARKS: **30**	PORTIONS : 1^{st} & 2^{nd} Module

Answer Scheme Marks: **15x2=30**

Q. NO	Answers	Marks
1.a	Given the set of FD's, F: { A→B, C→B, D → ABC, AC→ D} Step 1: Decomposing Functional Dependencies A→B, C→B, D → A, D →B, D →C, AC→ D	**1**
	Step 2: Check for Redundant Functional Dependencies i. A→B is ignored and check for redundant functional dependencies weather A can determine B checking for A closure i.e. A^+ = A hence A→B is not Redundant. ii. C→B is ignored and check for redundant functional dependencies weather C can determine B checking for C closure i.e. C^+ = C hence C→B is not Redundant. iii. D → A is ignored and check for redundant functional dependencies weather D can determine A checking for D closure i.e. D^+ = DBC hence D→A is not Redundant. iv. D → B is ignored and check for redundant functional dependencies weather D can determine B checking for D closure i.e. D^+ = DABC hence D→B is Redundant hence its removed. v. D → C is ignored and check for redundant functional dependencies weather D can determine C checking for D closure i.e. D^+ = DAB hence D→C is not Redundant. vi. AC→D is ignored and check for redundant functional dependencies weather AC can determine D checking for AC closure i.e. AC^+ = ACB hence AC→D is not Redundant. We are left with non-redundant functional dependencies A→B, C→B, D → A, D →C, AC→D.	**2**
	Step 3: We should solve Left hand side of Functional Dependencies A→B, C→B, D → A, D →C, AC→ D. here in our Functional Dependencies only AC→D is double in left hand side hence we need to solve AC→D Now find weather in C+ can we find A. Then A can be removed from above FD C^+= CB hence A is not present in C closure Now find weather in A+ can we find C. Then C can be removed from above FD A^+= AB hence C is not present in A closure Hence we can't further solve AC→D.	**2**
	So the minimal cover of F: { A→B, C→B, D → ABC, AC→ D} is **{A→B, C→B, D →AC, AC→ D}**	**2**
1.b	i) List the students' names who have enrolled for the 2nd semester. SELECT DISTINCT name FROM STUDENT s JOIN ENROLL e ON s.regno = e.regno WHERE e.sem = 2;	**2**
	ii) List the number of books adopted in semester '4'. SELECT COUNT(*) AS num_books_adopted FROM BOOK_ADOPTION WHERE sem = 4;	**2**
	iii) List the courses where the course name begins with 'a' and is held by the 'cs' department. SELECT c.course_id, c.cname FROM COURSE c WHERE c.cname LIKE 'a%' AND c.dept = 'cs';	**2**

	iv) List departments that have more than 2 courses. SELECT dept FROM COURSE GROUP BY dept HAVING COUNT(*) > 2;	**2**
2.a	Consider the relation R (A, B, C, D, E, F, G, H) with the following sets of FD's {A→BD, B→C, E→FG, AE→H} i. Closure $A\cancel{BCD}E\cancel{FGH}^{+}$ = ABCDEFGH AE^{+} = AEBDCFGH Proper Subset A^{+} = ABDC E^{+} = EFG CK = AE Prime Attributes Are A, E ii: {A→BD, B→C, E→FG, AE→H} Partial Dependencies Are A→BD, E→FG Hence the highest Normal Form is 1NF iii: find Closure of Partial Dependencies creating FD's i.e. 1. A^{+} = ABCD 2. E^{+} =EFG To be in Lossless Join Decomposition R_1= {A, B, C, D} R_2= {E, F, G} R_3= {H, A, E} Where A, E can drive R_1, R_2, Hence taken as common. Finding FD's of R_1, R_2, R_3 from FD's {A→BD, B→C, E→FG, AE→H} to check Dependency Preserving Decomposition. F_1={A→BCD, B→C} F_2={E→FG} F_3= {AE→H} CK = A, CK= E CK=AE A is Super Key E is Super Key AE is Super Key it's in BCNF it's in BCNF it's in BCNF B→C is Transitive Dependency iv: Now solve B→C so convert it into BCNF Consider R_1= {A, B, C, D} F_1={A→BCD, B→C} B^{+}= B, C R_{11}= {B, C} R_{12}={A, D, B} As there are 2 Attributes so R_{11} It's in BCNF. F_{12}= {A→BD} CK = A A is Super Key it's in BCNF Now Relation R_2, R_3, R_{11}, R_{12} are Sub-relations are required to convert Relation R to BCNF.	**1** **2** **2** **2** **1**
2.b	i. List customers with loans at 'MSRIT' branch: SELECT DISTINCT c.customer-name FROM Customer c JOIN Borrower b ON c.account-number = b.account-number JOIN loan l ON b.loan-number = l.loan-number WHERE l.branch-name = 'MSRIT' ORDER BY c.customer-name ASC; ii. Find customers with "yesvanthpur" in address: SELECT DISTINCT Customer.customer-name FROM Customer WHERE Customer.customer-street LIKE '%yesvanthpur%'; iii. List the loan numbers in ascending order of amount. SELECT loan-number FROM loan ORDER BY amount ASC;	**2** **2** **1**

	iv. Find the average loan amount at each branch. SELECT branch-name, AVG(amount) AS average_loan_amount FROM loan GROUP BY branch-name;	**2**
3.a	**Definition.** A **functional dependency**, denoted by $X \rightarrow Y$, between two sets of attributes X and Y that are subsets of R specifies a *constraint* on the possible tuples that can form a relation state r of R. The constraint is that, for any two tuples $t1$ and $t2$ in r that have $t1[X] = t2[X]$, they must also have $t1[Y] = t2[Y]$. This means that the values of the Y component of a tuple in r depend on, or are *determined by*, the values of the X component; alternatively, the values of the X component of a tuple uniquely (or **functionally**) *determine* the values of the Y component. We also say that there is a functional dependency from X to Y, or that Y is **functionally dependent** on X. The abbreviation for functional dependency is **FD** or **f.d.** The set of attributes X is called the **left-hand side** of the FD, and Y is called the **right-hand side**. Thus, X functionally determines Y in a relation schema R if, and only if, whenever two tuples of $r(R)$ agree on their X-value, they must necessarily agree on their Y-value. Inference Rules are as follows: 1. Reflexivity: X→X Then X→Y, $Y \subseteq X$ This always holds true because of Trivial Where X is a primary Key. 2. Transitivity: if (X→Y & Y→Z) Then X→Z its same as Ram is sibling of sham and Sham is sibling of Mohan So. If Name → Marks And Marks → Department so Name → Department. 3. Augmentation: if X→Y then XA→YA Where R.no →Name then another attribute can be added both side of dependencies (R.no, Marks → Name, Marks) 4. Union: if X→Y & X→Z then X→YZ If R.no →Name & R.no→Marks then R.no → Name, Marks. 5. Decomposition/ Splitting: if X→YZ then X→Y & X→Z can be done because X can individually determine Y and as well as Z or any one attribute at a time. e.g. Name, marks→ Dept., Course then Name, Marks → Dept. and Name, Marks → Course but Name → Dept., Course and Marks → Dept., Course is false can't be done. 6. Pseudo Transitivity: if X→Y & YZ→A) the XZ→A If R.no→Name & Name, Marks →Dept. then R.no, Marks → Dept. 7. Composition: if X→Y & A→B then XA→YB	**1** **1** **1** **1** **1** **1** **1** **1**
3.b	 State diagram of a transaction.	**3**

	A transaction must be in one of the following states: • **Active**, the initial state; the transaction stays in this state while it is executing. • **Partially committed**, after the final statement has been executed. • **Failed**, after the discovery that normal execution can no longer proceed. • **Aborted**, after the transaction has been rolled back and the database has been restored to its state prior to the start of the transaction. • **Committed**, after successful completion. The state diagram corresponding to a transaction appears in Figure We say that a transaction has committed only if it has entered the committed state. Similarly, we say that a transaction has aborted only if it has entered the aborted state. A transaction is said to have **terminated** if it has either committed or aborted. A transaction starts in the active state. When it finishes its final statement, it enters the partially committed state. At this point, the transaction has completed its execution, but it is still possible that it may have to be aborted, since the actual output may still be temporarily residing in main memory, and thus a hardware failure may preclude its successful completion. The database system then writes out enough information to disk that, even in the event of a failure, the updates performed by the transaction can be re-created when the system restarts after the failure. When the last of this information is written out, the transaction enters the committed state. As mentioned earlier, we assume for now that failures do not result in loss of data on disk. A transaction enters the failed state after the system determines that the transaction can no longer proceed with its normal execution (for example, because of hardware or logical errors). Such a transaction must be rolled back. Then, it enters the aborted state. At this point, the system has two options: • It can **restart** the transaction, but only if the transaction was aborted as a result of some hardware or software error that was not created through the internal logic of the transaction. A restarted transaction is considered to be a new transaction. • It can **kill** the transaction. It usually does so because of some internal logical error that can be corrected only by rewriting the application program, or because the input was bad, or because the desired data were not found in the database.	**2** **1** **1**